

# Customer Order Insights & Delivery Tracker

Capstone Project — Week 5

*Azure DevOps CI/CD Automation*

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|---------------------------------------|------------------------|
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# WEEK 5 – AZURE DEVOPS PIPELINE SIMULATION

## Objective

In week 5 I learnt the basic workflow of ci/cd automation using azure devops. The goal was to automate the execution of the customer order analysis script and generate logs automatically.

Since azure services were not available during implementation, the workflow was simulated using google colab and sample yaml configuration.

## Tools used

- Azure devops
- Python
- Yaml

## Workflow

Push code to repository

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Pipeline starts automatically

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Install dependencies

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Run python analysis script

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Generate output logs

### **Sample python script**

```
print("customer order pipeline started")  
print("loading order data")  
print("processing delayed deliveries")  
print("analysis completed successfully")
```

→ This script simulates the execution flow of the pipeline.

### **Sample yaml pipeline**

```
trigger:  
  - main  
pool:  
  vmImage: 'ubuntu-latest'  
steps:  
  - script: |  
      pip install pandas numpy pyspark  
      python run_pipeline.py  
    displayName: 'run customer order analysis'
```

→ The yaml file defines the automation process for running the python script.

### **Expected output**

- Pipeline execution started
- Required packages installed
- Customer delay analysis completed

- Output logs displayed successfully

## **Conclusion**

This task helped in understanding the basic concept of ci/cd pipelines and how automation can be used in data engineering projects for running scripts and generating reports automatically.